

Health Information Toolkit Process Book

Tim Hickey

Literature Review

Colorectal cancer, also known as CRC, is one of the most commonly diagnosed cancers in men and women. This disease develops when abnormal cells grow uncontrollably in the colon and/or rectum. These cells initially begin as small polyps but may become cancerous over time. CRC is preventable and treatable if detected early enough through screenings. Screenings allow for the removal of polyps and can diagnose cancer at early stages, when treatment is more likely to have positive outcomes. Despite this, CRC screening rates remain below national goals, especially among less fortunate demographics.

Screening for CRC is recommended starting at around age 45 for patients at average risk. If the patient's family has a history of CRC, then an earlier age is recommended. CRC screenings should be incorporated into men's and women's preventive health routines as they age. Screening methods range from colonoscopy procedures—which some patients may feel are too invasive—to less invasive stool tests. Colonoscopies are considered the best available test because they enable the detection and removal of polyps. Stool tests are more accessible and convenient but may still require a follow-up colonoscopy depending on initial results.

Despite the effectiveness of CRC screenings, the number of people who undergo them remains low. Patients report that limited knowledge about guidelines, concerns about discomfort, shame and embarrassment, fear of results, and logistical difficulties discourage them from pursuing CRC screenings. Other barriers include rigid work schedules, lack of insurance coverage, limited access to screenings, and emotional resistance.

Primary care “safety-net” systems that serve low-income and medically underserved populations face unique barriers as well. Competing demands during short visits, lack of reminder systems, and limited capacity for follow-up appointments hinder the delivery of CRC

screenings. Preventive care is often not the highest priority when patients present with urgent health or social needs. Providers may also struggle due to limited access to culturally appropriate educational and informational resources.

CRC also has a familial component; patients with a family history of CRC or hereditary cancers are at significantly higher risk than those without. This raises issues of transparency within families. Some families are open about hereditary cancers, which can increase screening rates, while others are dismissive and downplay the risk. These differences often reflect interpersonal and cultural dynamics. Stigma also consistently emerges as a barrier to CRC screenings. Patients may feel embarrassed or uncomfortable about the colonoscopy process, which requires direct access to the rectum. They may also fear being judged about their bowel habits or family history of cancer. Cultural taboos further reinforce these barriers.

Opportunities for design in CRC treatment lie primarily in the media surrounding CRC. The creation of advertisements and infographics that normalize CRC and humanize those affected by it may encourage more people to pursue screenings and treatment. While designing media that is effective across all demographics would be ideal, a one-size-fits-all approach is unlikely to succeed. Therefore, it is important to design targeted media for specific demographics to maximize effectiveness.

CRC remains a major public health concern, but it is effectively treatable and preventable when identified early. The biggest issue with CRC screenings is not their effectiveness or reliability, but rather their accessibility and appeal across all demographics. Overcoming these barriers is essential, as doing so will immediately improve the health and outcomes of those at risk.

Literature Matrix pt. 1

Literature Matrix Worksheet

Group Members: Tyler Dye, Tim Hickey, Jessica Lee

Instructions: Each student in your group should summarize the article they read by filling out a row in the matrix below. After all rows are complete, review the columns together to identify recurring themes, barriers, and opportunities for design.

Article Title & Author	Key Findings	Barriers to Screening or Assessment	Communication Challenges	Stigma or Bias	Opportunities for Design
"Patients' self-reported barriers to colon cancer screening in federally qualified health center settings" <i>Meera Muthukrishnan, Lauren D Arnold, & Aimee S James</i>	CRC is a leading cause of cancer-related death in the US Evidence shows screening reduces CRC incidence & mortality	Only 60% of age-eligible adults are up-to-date on CRC screening 22.9% said they weren't due for screening, nor did their provider recommend it Common barriers included fear or worry about the procedure or outcome, financial challenges (lack of insurance or cost of testing), and logistic challenges like transportation & time Few said that screening was of low importance or mentioned discomfort with the procedure or colonoscopy preparation	Possible that HCP recommended screening, but the patients didn't remember or understand it completely or correctly	n/a...	Raising awareness for CRC through the internet Create a simple, inclusive test to raise urgency & awareness
What kinds of cancer screenings should men get as they age? Howard E. LeWine, MD	Colonoscopies can find early cancers, doctors can remove polyps before they can become cancerous Periodic full-body MRIs and CT scans because they aren't 100% effective and can lead to uncertainty	Some patients may not be able to receive a colonoscopy and must do stool testing instead. Patients may also be uncomfortable with the idea of receiving a colonoscopy	Those with a genetic mutation, medical condition linked to cancer, or a history of cancer within the family may actually benefit from an image scan and additional testing	Patients may be uncomfortable with someone entering their colon Patients that can't receive a colonoscopy must do a stool test in order to be tested for cancer	Normalization of receiving a colonoscopy Normalization of taking a stool test if receiving a colonoscopy isn't possible
Patient and provider barriers to colorectal cancer screening in the primary care safety-net	The article focused on a study that utilized four focus groups with 40 patients and one with primary care providers from an urban safety-net	Organizational/System-Level Barriers: No referral pathways for colonoscopy after positive FOBT. No treatment options available for uninsured	Lack of System-Level Feedback Loops: Providers and clinics struggled with securing follow-up	Not specifically explored in this article, however patient comments can be interpreted as inherent stigma/bias,	Team-based care, reminders and feedback, to promote screening rather than relying solely on providers.

Listed Authors/Publishers: A.S O'Malley M.D., M.P.H. a, E Beaton B.A. a, K.R Yabroff Ph.D, M.B.A. b, R Abramson M.D. c, J Mandelblatt M.D., M.P.H. a	clinic serving uninsured, low-income African-American adults aged 50+. Comments were thematically categorized into eight content areas: Primary care characteristics (36%) Procedural issues related to screening (16%) Knowledge (14%) Cost/insurance coverage (13%) Ordering of priorities (12%) Attitudes (5%) Information sources (2%) Perceptions of discrimination (2%)	patients. Procedural and Clinic-Level Barriers: Challenges in coordination and follow-up care. Knowledge and Awareness Gaps: Limited patient and provider knowledge about screening guidelines. Financial Barriers: Costs of screening and lack of insurance coverage. Competing Priorities / Attitudes: Patients not prioritizing screening amidst life's demands; general attitudes affecting adherence. Perceptions of Discrimination: Though a smaller portion (2%), participants noted fears of discrimination influencing access.	information-e.g., not knowing whether patients completed colonoscopies elsewhere, leading to documentation gaps and "chasing" paperwork. Insufficient Referral and Education: Many patients lacked provider recommendation or referral, limiting awareness and follow-through. Language, Communication, and Trust: In safety-net settings, cultural and linguistic mismatches, low health literacy, and distrust impede the effectiveness of screening discussions.	including the perception of discrimination and the belief Providers may unconsciously deprioritize uninsured or minority patients due to systemic biases or resource constraints, though not explicitly studied here. There is also mention of an internal bias between patients of low prioritization of the subject.	Organizational coordination systems, policy advocacy, educational outreach, cultural competence, stigma-reducing interventions, provider bias awareness tools.
New approaches to colorectal cancer screening Heidi Godman	The Shield blood test by Guardant Health looks for DNA fragments shed from CRC tumors Out of 7800 patients, Shield was successful at identifying CRC cases 83% of the time Shield only had a 13% success rate at finding polyps Shield gave a falsely positive result 10% of the time	Traditional screenings are unpleasant and time-consuming Shield isn't FDA approved yet Shield struggled to find precancerous growths If a stool test finds something suspicious, the patient will need to book a follow-up colonoscopy, which they will have to pay for as they aren't entirely covered by insurance If the Shield test finds something suspicious, the patient will need to book a follow-up colonoscopy	The Shield test isn't 100% effective and may result in a false positive Patients may not realize that they may have to take a follow-up colonoscopy depending on their blood test results	Shield test isn't covered by insurance and may encourage the patient to take a follow-up colonoscopy which isn't covered by insurance, making a proper screening harder to access if you have low income Patients uncomfortable with colonoscopies may still have to receive one if the Shield test yields suspicious results The only people who used the Shield test were over the age of 45	A Shield test designed for younger people Informing people that the Shield test is a last resort alternative for those uncomfortable with a colonoscopy or stool test
"Examining intrafamilial communication of colorectal cancer risk"	First-degree relatives (FDR) of probands with CRC may be at increased risk of CRC	Barriers included a lack of information, negative expectations about familial reaction, the	Individuals with CRC tend to misunderstand colonoscopy	Cultural barriers impacted the decision to disclose to FDRs	Encourage & require HCPs to actively speak about CRC

Literature Matrix pt. 2

status to family members and kin responses to colonoscopy: a qualitative study" Kaitlin M McGarragle, Crystal Hare, Spring Holter, Dorian Anglin Facey, Kelly McShane, Steven Gallinger, and Tae L Hart	and require colonoscopy A better understanding of the barriers surrounding colonoscopy and disclosure is needed	assumption that FDRs were aware of risk or already being screened, dysfunctional family dynamics, and cultural barriers Additionally, there were also negative perceptions about colonoscopy, healthcare provider-related factors, practical concerns, and again, a lack of information about CRC, risk, and colonoscopy	guidelines for FDRs, believe colonoscopy is a forbidden subject, and report not receiving advice from HCPs on how to talk to their relatives about colonoscopy	for some probands who discussed stigma surrounding the topic of cancer in general	Create a course or get HCPs to teach individuals how to communicate with their families on the topic of CRC and whether or not it stems from an FDR
Patient-Reported Barriers to Colorectal Cancer Screening: A Mixed-Methods Analysis Listed Authors/Publisher s: Resa M. Jones PhD, MPH a c , Kelly J. Devers PhD e, Anton J. Kuzel MD b, Steven H. Woolf MD, MPH a b d	The article focuses on patients from a 2005 mixed-methods study involving primary care patients (age 50-75), the authors used a mailed survey and open-ended questions to explore the most significant barriers to CRC screening. Survey replies highlighted that fear (especially concerning the procedure and bowel prep) and the bowel preparation itself were the leading perceived barriers.	Fear and Prep-Related Concerns: Widespread anxiety about the colonoscopy procedure and its preparation Information and Time Constraints: Patients highlighted a lack of clear information and difficulty fitting screening into busy lives Lack of Physician Engagement: Many patients reported that their providers neither recommended nor adequately discussed screening Psychosocial & Cultural Dimensions: - Feelings of low self-worth - Para-sexual sensitivities (discomfort due to perceived sexual stigma)- Fatalism-viewing cancer as inevitable and screening as pointless- Negative past experiences with medical tests Skepticism about financial incentives from providers or the healthcare system	Insufficient Recommendations from Providers: Patients frequently reported that screening wasn't discussed or encouraged by their physicians Failing to address emotional/cultural barriers: Personal concerns of patients were not properly addressed or had feedback, suggesting that communication formats failed to engage patients' specific concerns effectively	"Para-sexual" Sensitivities: Some patients perceived colonoscopy as embarrassing or violating-involving intimacy in a stigmatized context Fatalism and Low Self-Worth: A sense of inevitability around cancer, or a belief that one "doesn't matter enough" to warrant screening Distrust or Skepticism Toward Motives: Patients questioned whether providers were recommending screening for financial gain rather than health benefits	Tailored communication tools addressing common emotional barriers. Normalizing prep anxiety, reducing stigma via relatable narratives or patient testimonials. Clear, transparent explanations about the purpose of screening, risks, benefits, and what to expect during prep and procedure. Simplified prep instructions, support channels (e.g., hotlines), and options for prep assistance to reduce anxiety and logistical friction.

After completing the matrix, answer these reflection questions as a group:

- What are the top 3 recurring barriers across the articles?

1. Financial issues
2. Lack of information
3. Initiation between both parties

- What is one insight that surprised you or challenged assumptions?

Insurance doesn't cover all costs for a follow-up colonoscopy.

- Which themes feel most relevant to design?

Presentation & Engagement

- What opportunities or directions do you see for your toolkit project?

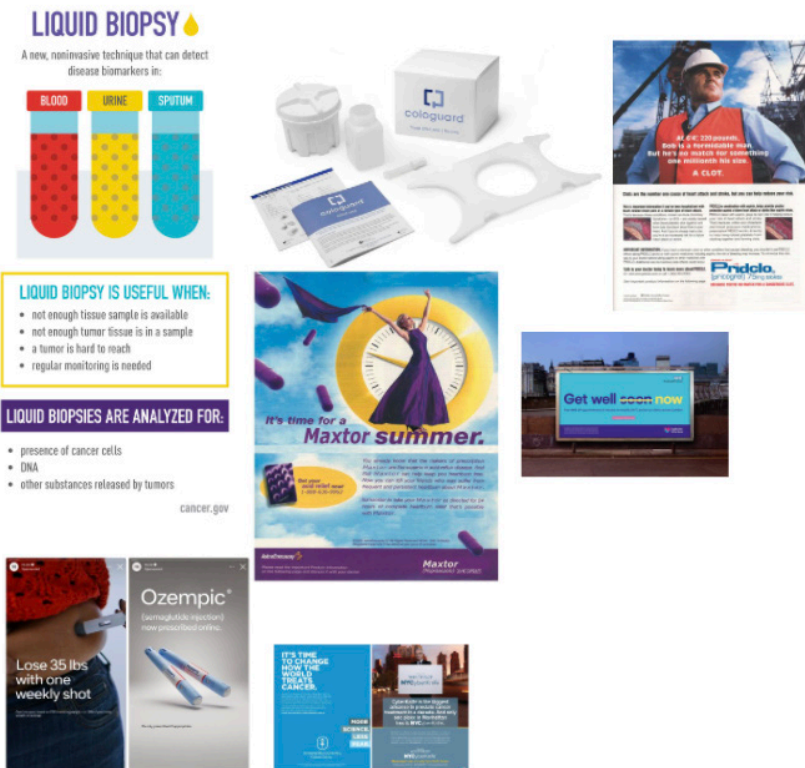
Advertisements to spread awareness

Stool/blood test kits that aren't intimidating

Reminder system/communicative notification system

Visual Campaign Audit

Tim



Discuss in your groups: Tyler, Tim, & Jessica Lee

What is the **tone** of the campaign?

- Imperative tone
- Casual suggestions
- Serious or casual (almost comical)
- Tries to convey its message "normally"

•What kind of **visuals** are they using?

- Usage of humans
- Lots of bright colors (blue or yellow)
- Medical colors??

•What **language** is used to build trust?

- Infographic; formal & informational
- Relatability

Branding Drafts

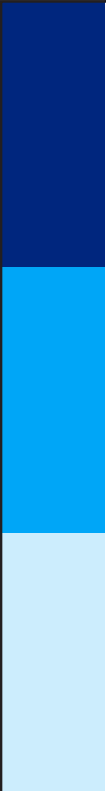


Eurostile Regular

Remedy/ColoRemedy



Keywords: Human, reliable, safe, professional





ColonAssist



ColonAssist



ColonAssist

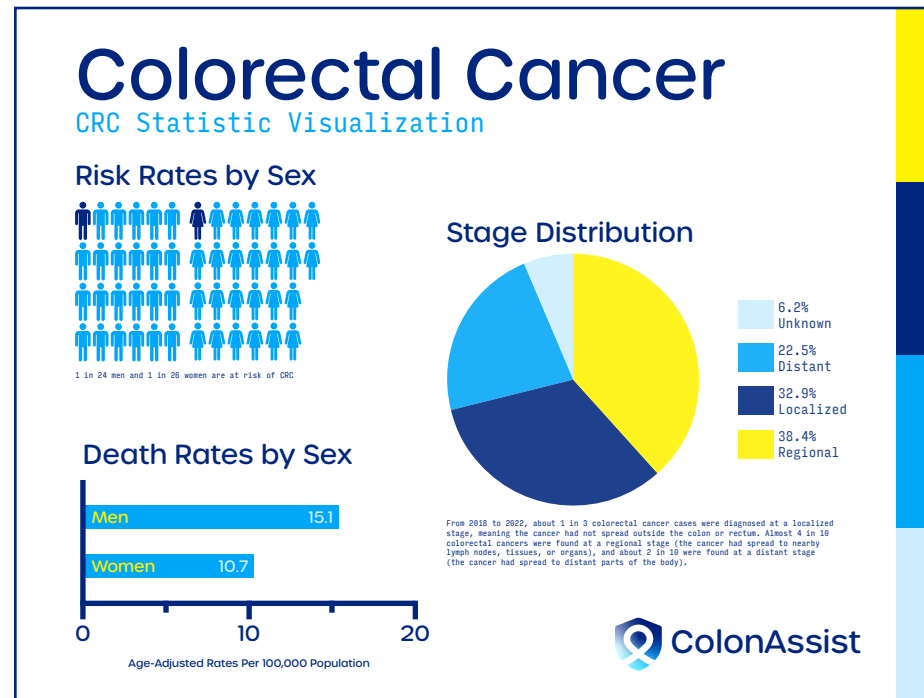
Branding Attributes

Adjectives + Explanations

- Human
 - Colon cancer is a human experience, its normal to be concerned about it
- Reliable
 - Trust is important in the health world, and it's important to be reliable when people place their trust in you
- Safe
 - If something is unsafe, people will be scared of it, by being safe, people will be more comfortable
- Professional
 - It's important to not be too casual as then people won't take you seriously, so professionalism is important

This brand aims to connect with people at a human level in order to make them more comfortable with treating and preventing colon cancer. Trust with patients is earned, and trust is the most important thing to have in the world of health.

Data Visualization



Content Outline

User-facing:

- To feel more comfortable discussing and pursuing colon cancer screenings/treatment
- To feel safe
- To trust the tool kit
- To not feel inconvenienced
- Reliable way to schedule a CRC screening
- Reliable source of information regarding CRC
- Reliable way to get screened for CRC

Physician-facing:

- To make the toolkit straightforward
 - If the user needs clarification, make it easily accessible
- To make the toolkit convenient
- Easy communication with patient
- To make the patient comfortable and not intimidated

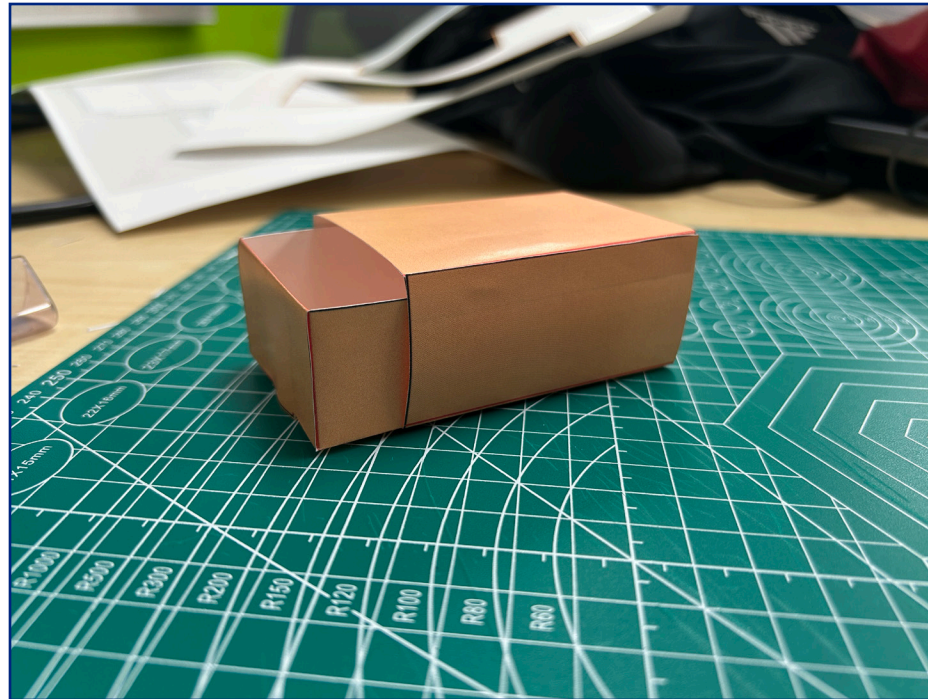
Content:

- Digital resource used for information about CRC
- FAQ about CRC
- Way to order a stool test or schedule a colonoscopy
- Stool test (if ordered)

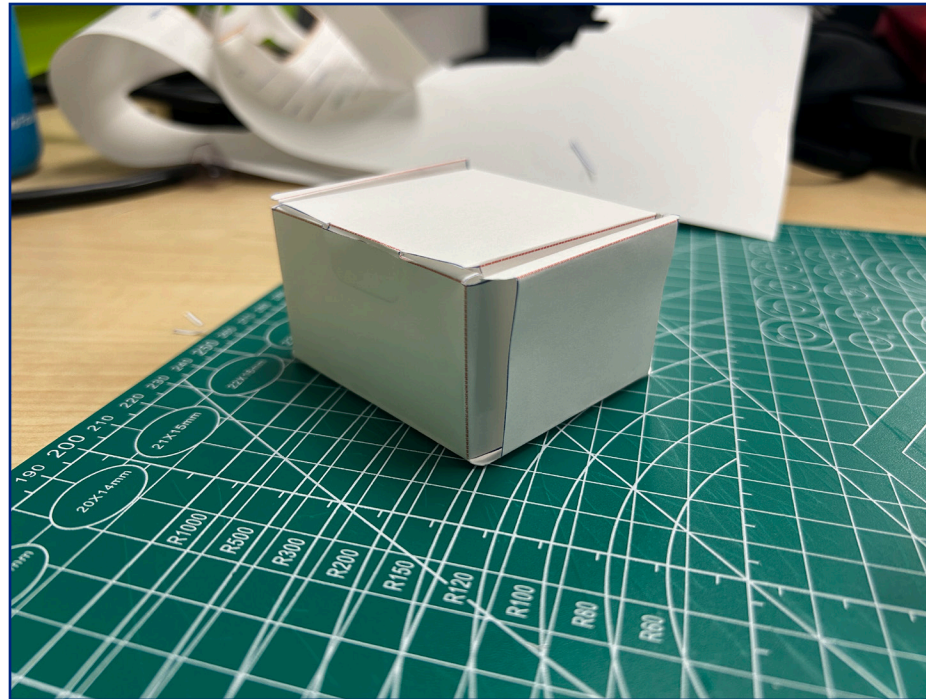
Mini Packaging pt. 1



Mini Packaging pt. 2



Mini Packaging pt. 3



Packaging Prototype



Written Reflection

When working on this project, my goal was to make colorectal cancer as approachable as possible. Considering that cancer is life-threatening, it's important to raise awareness about the less-discussed types of cancer. Since colorectal cancer (CRC) was an available topic for this project, I decided to focus on it.

When creating my designs, I wanted to avoid evoking fear or aggression, so I chose a color scheme featuring a gradient of blue and a single shade of yellow. This palette felt especially appropriate because blue is the official color for CRC. Blue is also a calming, non-threatening color, which I felt would work well for my branding.

For the logo, I designed a shield with a ribbon shaped like a person in the center. The shield symbolizes protection, while the ribbon/person represents colon cancer awareness and the protection of patients. Together with the color palette, the logo conveys a sense of calm and safety. These are qualities that help patients feel more comfortable discussing colon cancer.

Overall, I'm very happy with how this toolkit and branding turned out. I think the color scheme makes the toolkit look professional, and I'm particularly pleased with how the packaging came out. If I could change one thing, I would probably revise the contents of the toolkit by adding some small trinkets or fun items, such as stickers or a pen or pencil. These items would help the patient feel more connected and feel more comfortable discussing CRC as a result.